

COMICCRAFT

AI Comic Story Creator using Gemini Models

BRAINSTORMING & IDEATION PHASE

A structured ideation document describing the project concept, user needs, creative workflow, AI-assisted approach, and initial feature ideas for ComicCraft.

1. Introduction to Brainstorming & Ideation

The brainstorming and ideation phase establishes the initial concept behind ComicCraft: a web-based application that helps users transform a simple creative prompt into a personalized comic story with illustrations. The project documentation describes the application as combining AI-generated storylines, dialogues, scene descriptions, and comic-style images within a single workflow.

The central idea is to make comic creation accessible to non-artists, storytellers, and casual users by reducing the amount of manual effort required for writing, visual planning, illustration, and document creation.

2. Problem Identification

The project concept focuses on simplifying the comic-creation process. A user may have an idea for a story but still need to create the storyline, dialogues, scene descriptions, illustrations, panel arrangement, and final document separately. ComicCraft proposes a unified application that connects these activities.

Challenge identified	Ideation response in ComicCraft
Turning a simple idea into a structured story	Generate a panel-by-panel comic outline using AI
Writing narration and character dialogue	Use an AI storytelling model to expand the outline
Creating illustrations for each scene	Generate comic-style images from image prompts
Maintaining a connected panel sequence	Build a structured comic layout
Preparing a shareable final document	Export the completed comic as a PDF
Making the process accessible to non-artists	Provide a simple web-based input and preview interface

3. Core Idea Generation

The core idea developed for the project is an AI-powered comic story creator where the user provides a small set of creative inputs and the system produces a complete comic. The documentation identifies the main inputs as story prompt, character name, setting, story tone, and art style.

- **Story Prompt:** the main idea or situation that the comic should describe.
- **Main Character Name:** identifies the hero or central character.
- **Setting:** defines where the story takes place, such as a forest, school, city, or space.

- Story Tone: controls the mood, such as dramatic, funny, poetic, or light-hearted.
- Art Style: specifies the desired visual style, such as anime, pixel art, comic book, or realistic.

4. Creative Workflow Ideation

Ideation stage	Concept developed
User idea	User enters a creative story prompt and preferences.
Story planning	AI creates a structured five-panel outline.
Story expansion	AI develops narration and character dialogues.
Visual planning	Each panel receives an image-generation prompt.
Illustration	Stable Diffusion creates comic-style images.
Composition	Generated images and story text are matched into a comic layout.
Review	User views the complete comic through the preview page.
Final output	The comic is compiled into a downloadable PDF.

5. AI Technology Ideas Considered

The ideation process includes the use of separate AI capabilities for different creative tasks. The project documentation identifies Gemini Flash, Gemini Pro, and Stable Diffusion as the selected models for structured outline generation, detailed story generation, and comic illustration respectively.

Technology	Ideation purpose	Role in ComicCraft
Gemini Flash	Fast structured generation	Creates the five-panel comic outline
Gemini Pro	Detailed creative text generation	Creates narration and character dialogue
Stable Diffusion	Prompt-based visual generation	Creates comic-style illustrations
FastAPI	Application/backend framework	Connects user input, AI functions, routes, and templates
Jinja2	Dynamic web rendering	Displays user input results and generated comic content
FPDF	Document generation	Compiles comic panels and text into PDF

6. Example Brainstorming Scenario

The documentation provides the example idea: “A brave fox exploring an enchanted forest.” This illustrates how a short creative concept can be expanded into a complete personalized comic.

Creative element	Example idea
Story concept	A brave fox exploring an enchanted forest
Character	Fox
Setting	Enchanted forest
Tone	Dramatic
Art style	Anime
Expected result	A multi-panel comic containing story text and generated illustrations

7. Alternative Idea: Iterative Customization

A second documented scenario extends the idea by allowing users to change the tone and art style and regenerate the comic. For example, a user can select a funny tone and comic-book art style instead of previous choices. The system then regenerates the outline, story, and images using instructions tailored to the new preferences.

- Initial idea is entered by the user.
- The first comic is generated and reviewed.
- The user changes creative preferences.
- The generation pipeline is run again with the updated preferences.
- The revised comic can be reviewed and exported.

8. Feature Ideas Generated During Ideation

Feature idea	Purpose
Personalized comic prompts	Allow each user to control the story concept.
Multiple story tones	Support different moods and narrative styles.
Multiple art styles	Allow users to influence the visual appearance.
Five-panel structured output	Provide a consistent comic-generation format.
Panel-by-panel preview	Let users review the generated storyline and images.
Image prompt reference	Expose the artistic description associated with a panel.
PDF export	Provide a portable and shareable final comic.
Image testing route	Allow developers to test image generation independently.
JSON API generation	Support programmatic interaction with comic generation.

9. Ideation of System Architecture

The documentation organizes the application into three primary components: Frontend, Backend, and AI Integration. This separation provides a clear conceptual structure for the idea before implementation.

Frontend

HTML, CSS, and Jinja2 provide the user interface for collecting creative inputs and displaying generated comics.

Backend

FastAPI manages routing, receives form or JSON input, coordinates the generation workflow, and handles export.

AI Integration

Gemini APIs provide story generation while Hugging Face Diffusers and Stable Diffusion provide illustrations.

10. User Experience Ideas

- Keep the initial input form simple and focused on the most important creative choices.
- Provide a direct flow from input submission to generated comic preview.
- Present each comic panel clearly and sequentially.
- Allow the user to review the generated story before final export.
- Provide an explicit success page after PDF creation.
- Allow the user to return to the homepage and create another comic.

11. Future Ideas Identified in the Documentation

The project documentation states that the current version focuses on single comic generation without user account management. It also identifies possible future enhancements enabled by the modular architecture.

- User profiles
- Comic libraries
- Multi-page story arcs

12. Ideation-to-Development Transition

The brainstorming concepts naturally lead into the documented project milestones. The project moves from model selection and architecture, to core functionality development, route development, frontend development, and local deployment. This transition turns the initial creative idea into an implementable web application.

Milestone	Connection to the ideation
1. Model selection and Architecture	Select AI capabilities and define how frontend, backend, and AI services interact.
2. Core functionalities Development	Implement outline generation, story generation, image generation, layout, and PDF
3. routes.py Development	Connect user interactions and AI workflows through FastAPI routes.

Milestone	Connection to the ideation
4. Frontend Development	Turn the user-experience ideas into input, preview, and success pages.
5. Deployment	Run and verify the complete concept locally.

13. Brainstorming & Ideation Conclusion

The brainstorming and ideation phase establishes ComicCraft as an AI-assisted creative platform that connects story ideation, structured comic planning, narrative generation, visual generation, preview, and PDF export. The documented scenarios show how a short prompt and a small set of creative preferences can serve as the starting point for a complete personalized comic.

The resulting idea is modular and supports later expansion. The documentation specifically identifies user profiles, comic libraries, and multi-page story arcs as possible future enhancements.

Source basis: This document is derived from the uploaded ComicCraft project documentation. Where the source describes an implemented feature, it is presented as part of the documented project; future enhancements are presented only as future ideas identified in that documentation.